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Field Avare Biotech App Validation powered by MAKSA technology on Frozen-Thawed Bovine Semen (Okara, Pakistan, 2026)

Date: January 2026
Location: Okara, Pakistan

Conducted by: **Faseeh Ahmad**	Verified by: Avare Biotech Team (Dubai, UAE)
Sex Semen Technologist at GBT Pvt Ltd (Stgenetics Licensed Laboratoty, USA)	Maksim Mershiev, The CEO & Co-Founder of Avare Biotech
Signature: Signed by: 3D8F1A5F10B84FD... 1/19/2026	Signature: Signed by: Maksim Mershiev 325996BF0B2B43A... 1/18/2026

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Executive Summary / Conclusions

- Smartphone-based **MAKSA technology demonstrated ease of use** in the analysis of frozen–thawed bull semen, and the results were found to be **reliable and consistent** for routine semen quality assessment
- **Progressive motility values obtained through MAKSA were consistently higher** compared to those measured by conventional CASA systems, which may be attributed to **differences in tracking algorithms, frame rate, and field selection criteria**
- Overall, MAKSA represents a **cost-effective, portable alternative** to CASA systems, especially beneficial for **on-farm semen evaluation.**

Objective

To validate the smartphone-based MAKSA (Mobile-Assisted Key Semen Analysis) technology powering Avare Biotech app for **Frozen and Fresh** semen evaluation against conventional CASA and Nucleocounter methods.

Materials and Methods

The present study was conducted to evaluate the performance of a smartphone-based MAKSA technology in comparison with a conventional Computer-Assisted Semen Analysis (CASA) system for the analysis of frozen–thawed bull semen. 4 frozen semen straws were collected from proven bulls maintained at a commercial semen production unit.

Semen Thawing Procedure

Frozen semen straws were thawed in a water bath maintained at **37°C for 45 seconds**. After thawing.

Smartphone-Based MAKSA Analysis

Semen evaluation using MAKSA technology was performed with a smartphone mounted on a compound light microscope using a universal adaptor. A drop of thawed semen (5-7 Ml) was placed on a clean, pre-warmed glass slide and covered with coverslip. Videos were recorded using mobile at **3X** zoom under **microscope at 200× magnification**.

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Table 1 - MAKSA results comparison

Tests done	Key Semen Parameters	CASA	MAKSA	Difference	Difference, %
Bull 1-GUY Niliravi (Buffalo)	Concentration M/ml	**19.51**	**27.20**	**-7.69**	**-39.4**
	Motility %	**63.7**	**61**	**2.7**	**4.2**
	Progressive Motility %	**33.7**	**41**	**-7.3**	**-21.6**
Bull 2-TIGER (Holstein Friesian)	Concentration M/ml	**49.02**	**40**	**9.02**	**18.4**
	Motility %	**82.9**	**99**	**-16.1**	**-19.4**
	Progressive Motility %	**68.1**	**75**	**-6.9**	**-10**
Bull 3-BOREALIS (Holstein Friesian)	Concentration M/ml	**80.12**	**91.20**	**-11.08**	**-13.82**
	Motility %	**40.6**	**34**	**6.6**	**16**
	Progressive Motility %	**25.3**	**24**	**1.3**	**5.1**
Bull 4-S91/16 Niliravi (Buffalo)	Concentration M/ml	**9.51**	**20.80**	**-11.29**	**-118.7**
	Motility %	**63.7**	**80**	**-16.3**	**-25.5**
	Progressive Motility %	**50.5**	**59**	**-8.5**	**-16.8**

Discussion / Expert Insights

- Motility and progressive motility correlate well with CASA readings.
- The current version of the application does not support video capture using Phase contrast microscope to make it easier to use for majority of the specialists.
- Environmental factors such as light intensity, focus stability, and microscope settings had a noticeable impact on motility and concentration measurements, and I think it can be improved more.
- Proper sample handling, including dilution accuracy, temperature control, and timely analysis, was critical for obtaining reproducible results and usually simple compound microscopes do not have sperm slide temperature control facility.
- The app performed better for motility assessment than for concentration estimation, suggesting that algorithm refinement is particularly needed for sperm counting functions.
- Overall, the Avare Biotech MAKSA app shows strong potential as a rapid, field-based semen analysis tool, but requires further software training and technical refinement to match laboratory-grade CASA systems.

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Recommendations

- To obtain more accurate and reliable results, it is strongly recommended that videos be recorded from at least **5 different microscopic fields**, as each field under the microscope lens shows variation in both **progressive and total motility**.
- Also, I'd consider measuring over 200 sperm cells before giving results.
- The application should be trained in such a way that it analyzes multiple fields, calculates their average values, and then generates the result. This strategy will likely also help resolve the **concentration estimation issues** currently observed in the app

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Appendix

- Screenshot of the semen videos made and CASA results
- MAKSA PDF report samples

BULL 1: CASA Readings

Semen Production Unit

Qadirabad, Sahiwal

L&DD, Punjab

Animal Breeder Report

Animal ID: GUY

Analysis Date and Time: 09/01/2026

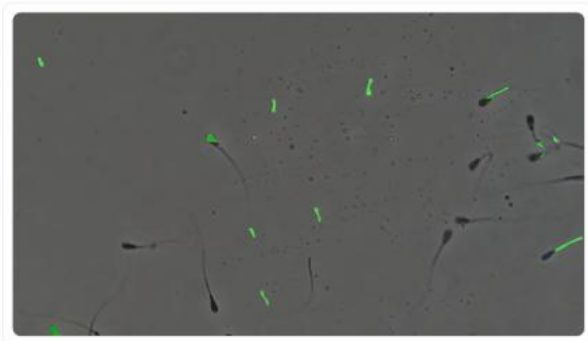
Motility

	Count	Sample M	Concentration M/ml	Percent Of Total
Total	204	10	19.51	100
Progressive	57	2	2.14	33.7
Motile	130	6	6.06	63.7
Slow	28	1	1.31	13.7

MAKSA Readings

Analysis 10.01.2026

Animal: — Name: Guy Age: 11 Breed: Nilliravi Status: Ready Created: 2026-01-10 02:34



Total objects:	17
Concentration (millions/ml^3):	27.20
Dilution rate:	[1:0]
Volume of field of view (mm^3):	0.002797
Progressive Motility:	41%
Non-Progressive Motility:	20%
Immotility:	39%

BULL 2

Semen Production Unit

Qadirabad, Sahiwal

L&DD, Punjab

Animal Breeder Report

Animal ID: Tiger

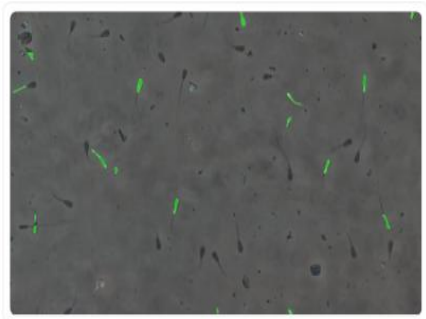
Analysis Date and Time: 09/01/2026

Motility

	Count	Sample M	Concentration M/ml	Percent Of Total
Total	389	10	49.02	100
Progressive	131	2	2.14	68.1
Motile	190	6	6.06	82.9
Slow	28	1	1.31	13.7

Analysis 10.01.2026

Animal: — Name: Tiger Age: 9 Breed: HF Status: Ready Created: 2026-01-10 01:54



Total objects:	25
Concentration (millions/ml^3):	40
Dilution rate:	[1:0]
Volume of field of view (mm^3):	0.001993
Progressive Motility:	75%
Non-Progressive Motility:	55%
Immotility:	1%

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BULL 3

Semen Production Unit

Qadirabad, Sahiwal

L&DD, Punjab

Animal Breeder Report

Animal ID: Borealis

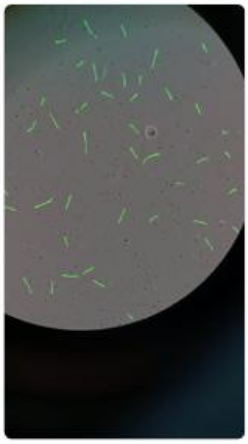
Analysis Date and Time: 09/01/2026

Motility

	Count	Sample M	Concentration M/ml	Percent Of Total
Total	349	58	80.12	100
Progressive	52	5	4.66	25.3
Motile	179	30	29.81	40.6
Slow	41	7	6.83	11.7

Analysis 10.01.2026

Animal: — **Name:** Borealis **Age:** 6 **Breed:** HF **Status:** Ready **Created:** 2026-01-10 01:23



Total objects:	57
Concentration (millions/ml^3):	91.20
Dilution rate:	[1:0]
Volume of field of view (mm^3):	0.006957
Progressive Motility:	24%
Non-Progressive Motility:	10%
Immotility:	66%

BULL 4

Semen Production Unit

Qadirabad, Sahiwal

L&DD, Punjab

Animal Breeder Report

Animal ID: S 91/16

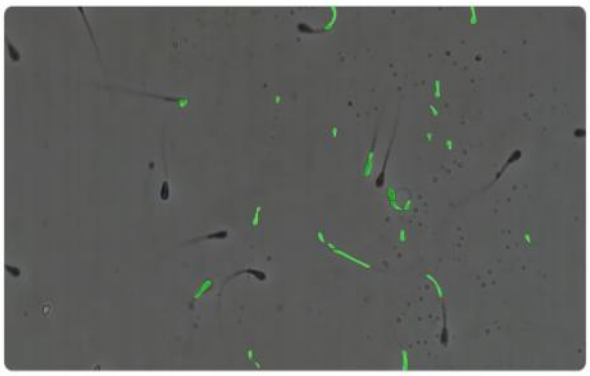
Analysis Date and Time: 12/20/2025 12:29:16

Motility

	Count	Sample M	Concentration M/ml	Percent Of Total
Total	204	10	9.51	100
Progressive	46	2	2.14	50.5
Motile	130	6	6.06	63.7
Slow	28	1	1.31	13.7

Analysis 31.12.2025

Animal: — **Name:** Buff **Age:** 8 **Breed:** Buff **Status:** Ready **Created:** 2025-12-30 19:44



Total objects:	13
Concentration (millions/ml^3):	20.80
Dilution rate:	[1:0]
Volume of field of view (mm^3):	0.002072
Progressive Motility:	59%
Non-Progressive Motility:	21%
Immotility:	20%